1. a) Expected number of indices where $i < a_i$.

$X_i = X(i < a_i)$

$X_i = \begin{cases} 1 & \text{if } i < a_i \\ 0 & \text{otherwise} \end{cases}$

$P(X_i) = \frac{n-i}{n}$ $\quad \frac{\text{total} - \text{index}}{\text{total}}$

$E(X) = \sum_{i=1}^{n} E(X_i)$

$= \sum_{i=1}^{n} \frac{n-i}{n}$

$= \frac{1}{n} \sum_{i=1}^{n} n-i$

$= \frac{1}{n}\left(\frac{(n-1)((n-1)+1)}{2}\right)$

$= \frac{(n-1)(n)}{2n}$

$= \frac{n-1}{2}$

b) Expected number of interior elements a_i such that $a_{i-1} < a_i < a_{i+1}$

$X_i = X(a_{i-1} < a_i < a_{i+1})$

$X_i = \begin{cases} 1 & \text{if } a_{i-1} < a_i < a_{i+1} \\ 0 & \text{otherwise} \end{cases}$

$P(X_i) = \frac{1}{3!} = \frac{1}{6}$ $\quad 3!$ ways to arrange a_{i-1}, a_i, a_{i+1}
1 way for $a_{i-1} < a_i < a_{i+1}$

Interior elements $= [2, n-1]$

$E(X) = \sum_{i=2}^{n-1} \frac{1}{6} = (n-1-1)\frac{1}{6}$

$= \frac{n-2}{6}$

c) Compute expected number of swaps

$X_i = X(a_i = j \text{ and } a_j = i)$

$X_i = \begin{cases} 1 & \text{if } a_i = j \text{ and } a_j = i \\ 0 & \text{otherwise} \end{cases}$

$P(X_i) = \frac{1}{n} \cdot \frac{1}{n-1}$ choose 1 out of n numbers to be in correct position; can only be 1 other number's swap out of n-1 other numbers

$= \frac{1}{n(n-1)}$

$E(X) = \binom{n}{2}\left(\frac{1}{n(n-1)}\right)$ choose 2 numbers to swap

$= \frac{n!}{2!(n-2)!\, n(n-1)}$

$= \frac{n \cdot (n-1) \cdot (n-2)!}{2\,(n-2)!\, n(n-1)}$

$= \frac{1}{2}$

2. a) Compute expected number of rooms without any paintings

$X_i = X(\text{Room } i \text{ has no paintings})$

$X_i = \begin{cases} 1 & \text{if room } i \text{ has no paintings} \\ 0 & \text{otherwise} \end{cases}$

$P(X_i) = \frac{11^9}{12^9}$

$E(X_i) = \sum_{i=1}^{12} \frac{11^9}{12^9} = 12 \cdot \left(\frac{11}{12}\right)^9$

$= 5.48$ rooms

Compute expected number of rooms with exactly 1 painting

$X_i = X(\text{Room } i \text{ has exactly 1 painting})$

$X_i = \begin{cases} 1 \text{ if room } i \text{ has exactly 1 painting} \\ 0 \text{ otherwise} \end{cases}$

$P(X_i) = 9 \cdot \frac{1}{12} \cdot \frac{11^8}{12^8}$

$E(X_i) = \sum_{i=1}^{12} 9 \cdot \frac{11^8}{12^9}$

$= 9 \cdot 12 \cdot \frac{11^8}{12^9}$

≈ 4.49 rooms

b) Compute expected number of rooms with no prints (indistinguishable)

$X_i = X(\text{Room } i \text{ has no prints})$

$X_i = \begin{cases} 1 \text{ if room } i \text{ has no prints} \\ 0 \text{ otherwise} \end{cases}$

$P(X_i) = \frac{\binom{9+11-1}{11-1}}{\binom{20}{11}} = \frac{\binom{19}{10}}{\binom{20}{11}}$

$\binom{9+12-1}{12-1} = \binom{20}{11}$
total ways to arrange prints

$\frac{\frac{19!}{10!\,9!}}{\frac{20!}{11!\,9!}} = \frac{19!\,11!\,9!}{10!\,9!\,20!} = \frac{11}{20}$

$E(X_i) = \sum_{i=1}^{12} \frac{11}{20}$

$= 12 \cdot \frac{11}{20}$

≈ 6.6 rooms